Zoology Legends

Zoo501

Quiz No.3 Spring 2023

Muhammad Zaman

0304-4756496

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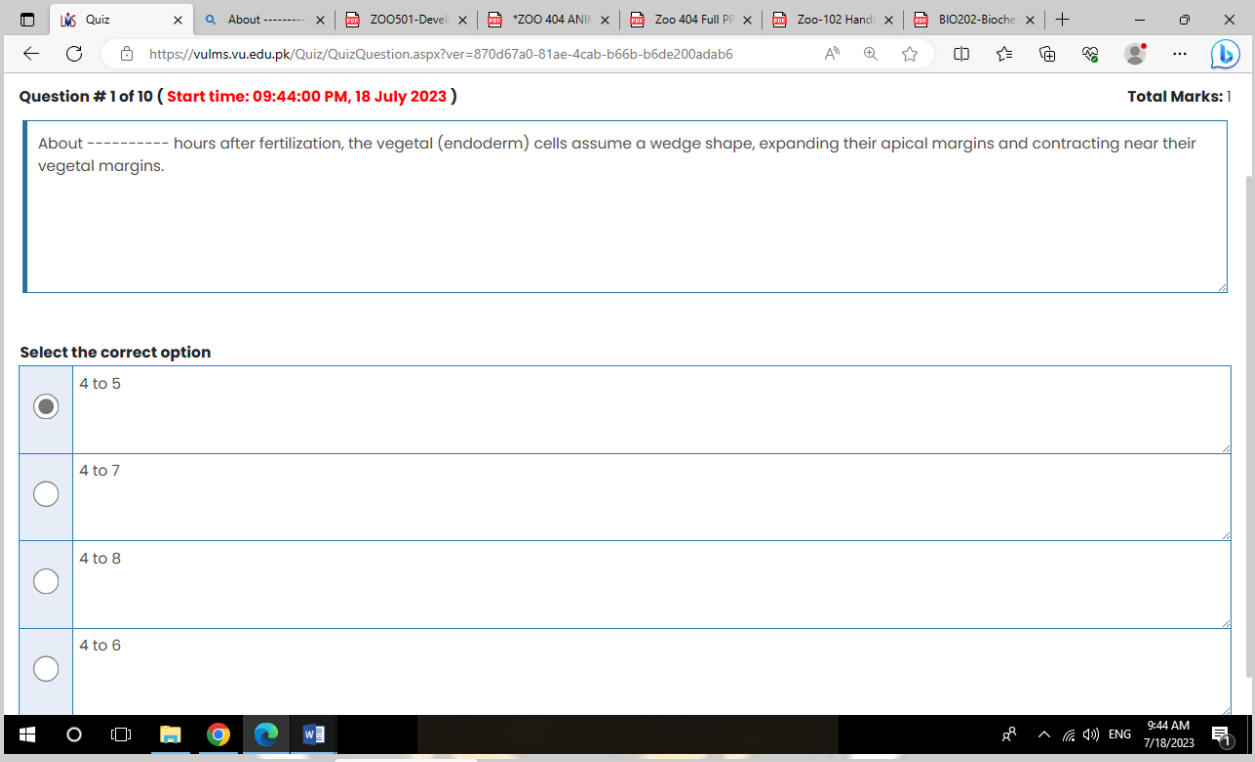
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Quiz | About --------- | ZOO501-Devel | *ZOO 404 ANII | Zoo 404 Full PF | Zoo-102 Hand | BIO202-Biochem

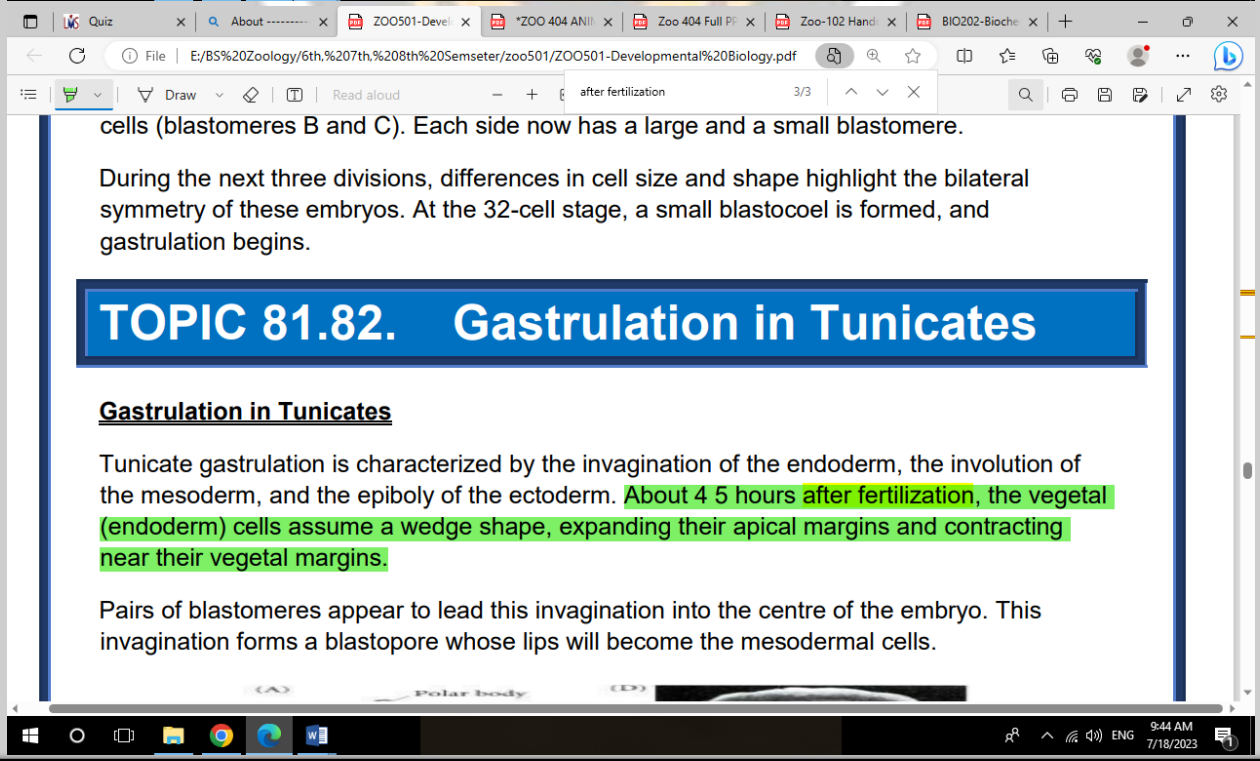
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Question # 1 of 10 (Start time: 09:44:00 PM, 18 July 2023) **Total Marks:** 1

About ---------- hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins.

Select the correct option

- (●) 4 to 5
- ( ) 4 to 7
- ( ) 4 to 8
- ( ) 4 to 6



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Draw | Read aloud | after fertilization 3/3

cells (blastomeres B and C). Each side now has a large and a small blastomere.

During the next three divisions, differences in cell size and shape highlight the bilateral symmetry of these embryos. At the 32-cell stage, a small blastocoel is formed, and gastrulation begins.

TOPIC 81.82. Gastrulation in Tunicates

Gastrulation in Tunicates

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm. About 4 5 hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins.

Pairs of blastomeres appear to lead this invagination into the centre of the embryo. This invagination forms a blastopore whose lips will become the mesodermal cells.

(A) Polar body (D)

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Quiz | Spiral holoblast | *ZOO501-Deve | *ZOO 404 ANII | Zoo 404 Full PF | Zoo-102 Hand | BIO202-Biochem

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Question # 2 of 10 (Start time: 09:44:30 PM, 18 July 2023) **Total Marks:** 1

Spiral holoblastic cleavage is characteristic of several animal groups like

Select the correct option

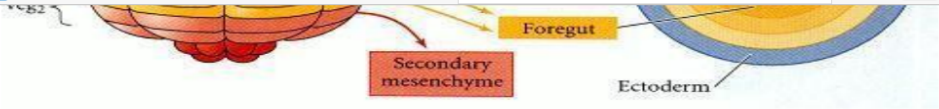
- ○ worms
- ○ arthropodes
- ◉ Molluscus
- ○ snakes

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Quiz | Spiral holoblast | *ZOO501-Deve | *ZOO 404 ANII | Zoo 404 Full PF | Zoo-102 Hand | BIO202-Biochem

File | E:/BS%20Zoology/6th,%207th,%208th%20Semseter/zoo501/ZOO501-Developmental%20Biology.pdf

Draw | Read aloud | Spiral holoblastic 1/1



TOPIC 75.76. Cleavage in Snail Eggs AND Sinistral coiling in Snail

The Early Development of Snails

Cleavage in Snail Eggs

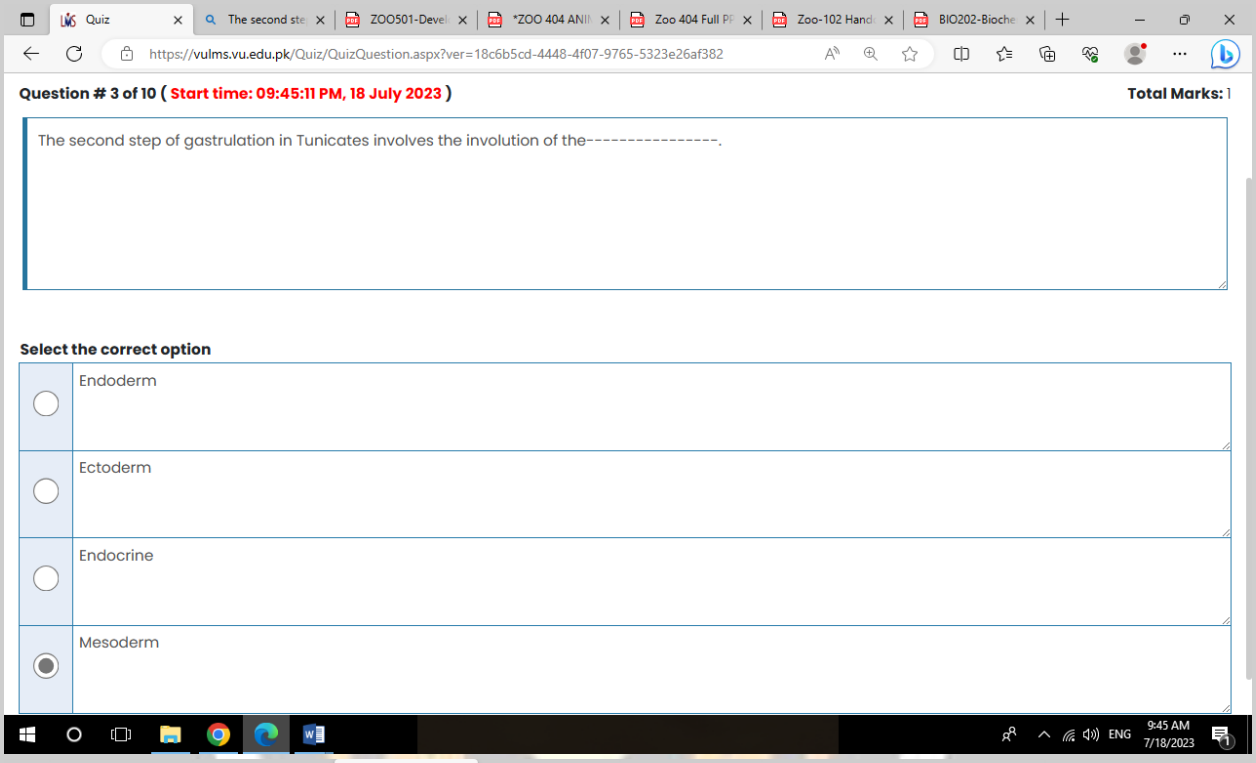
Spiral holoblastic cleavage is characteristic of several animal groups, including annelid worms, some flatworms, and most molluscs. It differs from radial cleavage in numerous ways..

MUHAMMAD IMRAN 45

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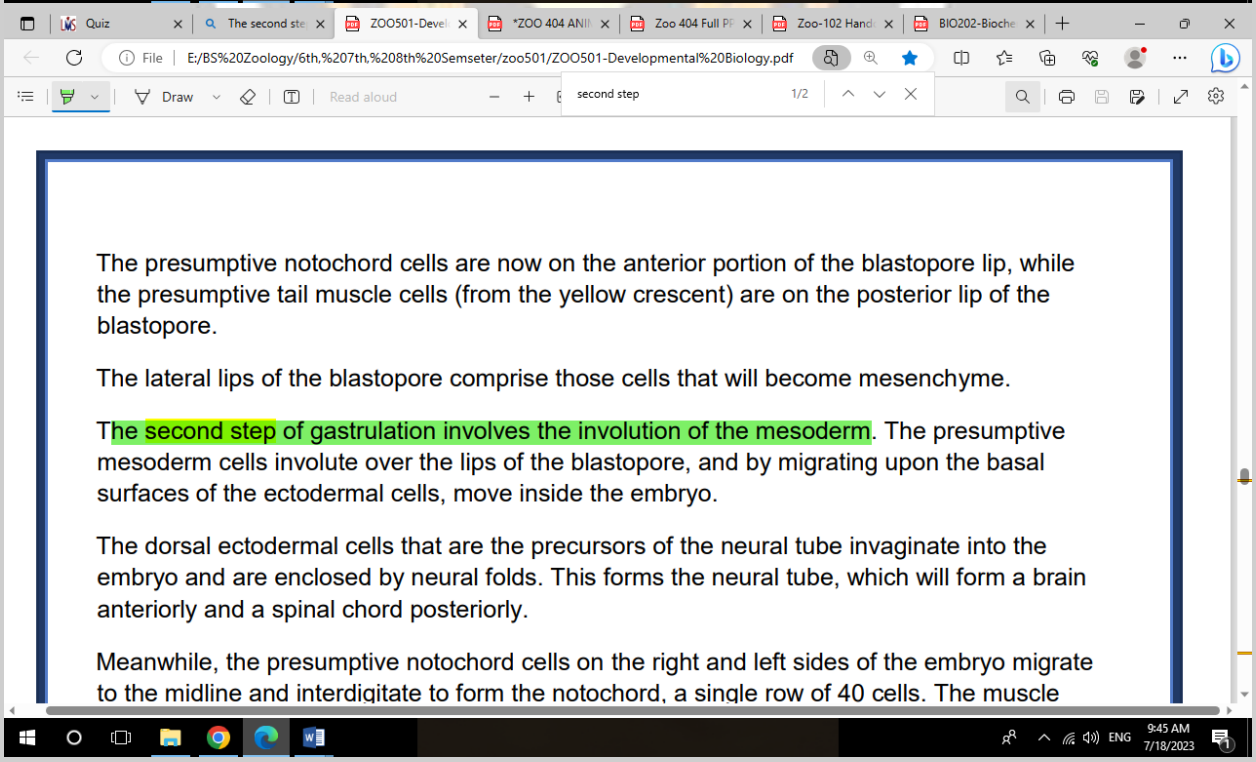


Question # 3 of 10 (Start time: 09:45:11 PM, 18 July 2023) Total Marks: 1

The second step of gastrulation in Tunicates involves the involution of the----------.

Select the correct option

- Endoderm
- Ectoderm
- Endocrine
- Mesoderm



The presumptive notochord cells are now on the anterior portion of the blastopore lip, while the presumptive tail muscle cells (from the yellow crescent) are on the posterior lip of the blastopore.

The lateral lips of the blastopore comprise those cells that will become mesenchyme.

The second step of gastrulation involves the involution of the mesoderm. The presumptive mesoderm cells involute over the lips of the blastopore, and by migrating upon the basal surfaces of the ectodermal cells, move inside the embryo.

The dorsal ectodermal cells that are the precursors of the neural tube invaginate into the embryo and are enclosed by neural folds. This forms the neural tube, which will form a brain anteriorly and a spinal chord posteriorly.

Meanwhile, the presumptive notochord cells on the right and left sides of the embryo migrate to the midline and interdigitate to form the notochord, a single row of 40 cells. The muscle

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Quiz | The snail stereo | *ZOO501-Deve | *ZOO 404 ANII | Zoo 404 Full PF | Zoo-102 Hand | BIO202-Biochem

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Question # 4 of 10 (Start time: 09:45:35 PM, 18 July 2023) **Total Marks:** 1

The snail stereoblastula is relatively ----------.

Select the correct option

- (●) Small
- ( ) Very large
- ( ) Tiny
- ( ) Large

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Quiz | The snail stereo | *ZOO501-Deve | *ZOO 404 ANII | Zoo 404 Full PF | Zoo-102 Hand | BIO202-Biochem

File | E:/BS%20Zoology/6th,%207th,%208th%20Semseter/zoo501/ZOO501-Developmental%20Biology.pdf

Draw | Read aloud | snail stereoblastula 1/1

- It is responsible for permitting the inductive interactions (through the material entering the 3d blastomere) leading to the formation of the shell gland and eye.
- It contains determinants needed for specifying the dorsal-ventral axis of the embryo.

TOPIC 77.78. Gastrulation in Snails AND Shell formation during gastrulation in snails

Gastrulation in Snails

The snail stereoblastula is relatively small.

Gastrulation in Snails is accomplished primarily by epiboly, wherein the micromeres at the animal cap multiply and "overgrow" the vegetal macromeres.

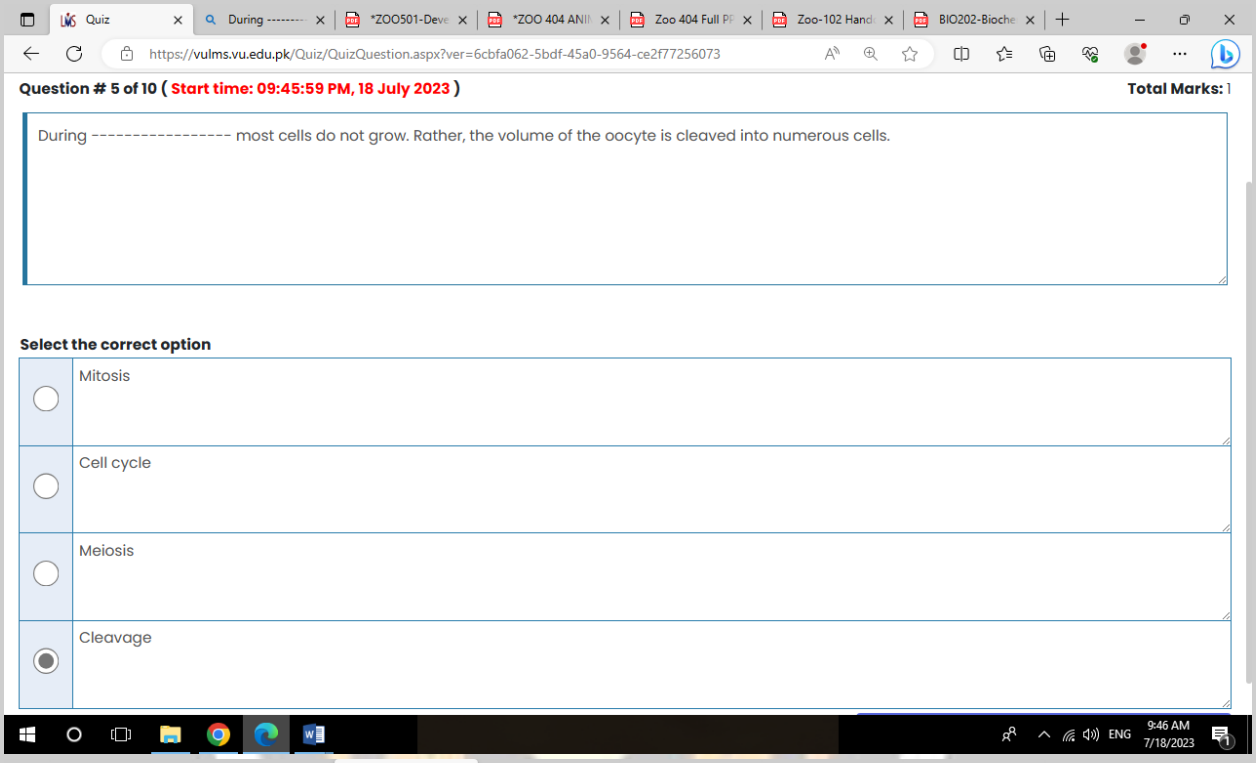
Eventually, the micromeres will cover the entire embryo, leaving a small slit at the vegetal pole.



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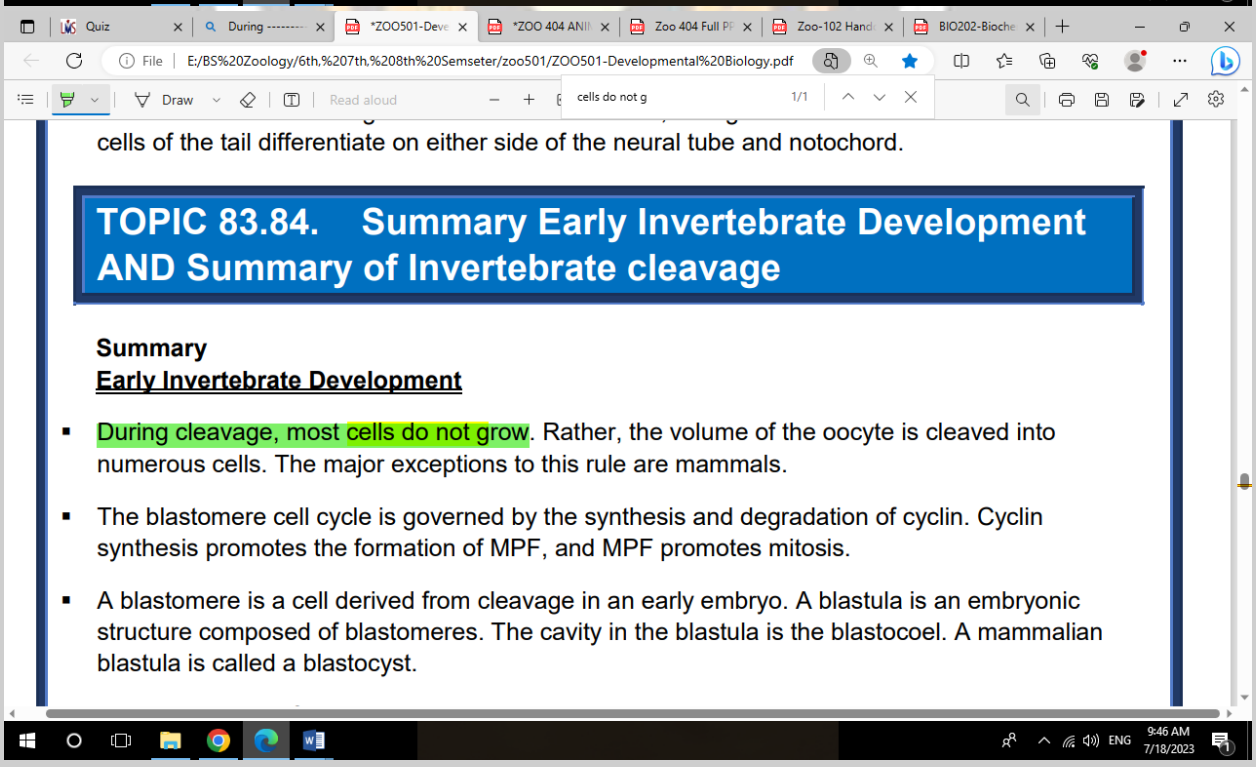


Question # 5 of 10 (Start time: 09:45:59 PM, 18 July 2023) Total Marks: 1

During ---------- most cells do not grow. Rather, the volume of the oocyte is cleaved into numerous cells.

Select the correct option

- Mitosis
- Cell cycle
- Meiosis
- Cleavage



cells of the tail differentiate on either side of the neural tube and notochord.

TOPIC 83.84. Summary Early Invertebrate Development AND Summary of Invertebrate cleavage

Summary

Early Invertebrate Development

- During cleavage, most cells do not grow. Rather, the volume of the oocyte is cleaved into numerous cells. The major exceptions to this rule are mammals.
- The blastomere cell cycle is governed by the synthesis and degradation of cyclin. Cyclin synthesis promotes the formation of MPF, and MPF promotes mitosis.
- A blastomere is a cell derived from cleavage in an early embryo. A blastula is an embryonic structure composed of blastomeres. The cavity in the blastula is the blastocoel. A mammalian blastula is called a blastocyst.

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Quiz | The cell division | *ZOO501-Deve | *ZOO 404 ANII | Zoo 404 Full PF | Zoo-102 Hand | BIO202-Biochem

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Question # 6 of 10 (Start time: 09:46:20 PM, 18 July 2023) **Total Marks:** 1

The cell divisions do not completely divide the egg, so this type of cleavage is called----------

Select the correct option

- (●) Mesoblast
- ( ) Blastocyte
- ( ) Mesentoblast
- ( ) Holoblast

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Quiz | The cell division | *ZOO501-Deve | *ZOO 404 ANII | Zoo 404 Full PF | Zoo-102 Hand | BIO202-Biochem

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Draw | Read aloud | divide the egg 1/1

Early Development of Vertebrates

Cleavage in Fishes

In fish eggs, cleavage occurs only in the **blastodisc**, a thin region of yolk-free cytoplasm at the animal cap of the egg. Most of the egg cell is full of yolk. The cell divisions do not completely divide the egg, so this type of cleavage is called **meroblastic.**

Since only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called **discoidal**. The calcium waves initiated at fertilization stimulate the contraction of the actin cytoskeleton to squeeze non-yolky cytoplasm into the animal pole of the egg.

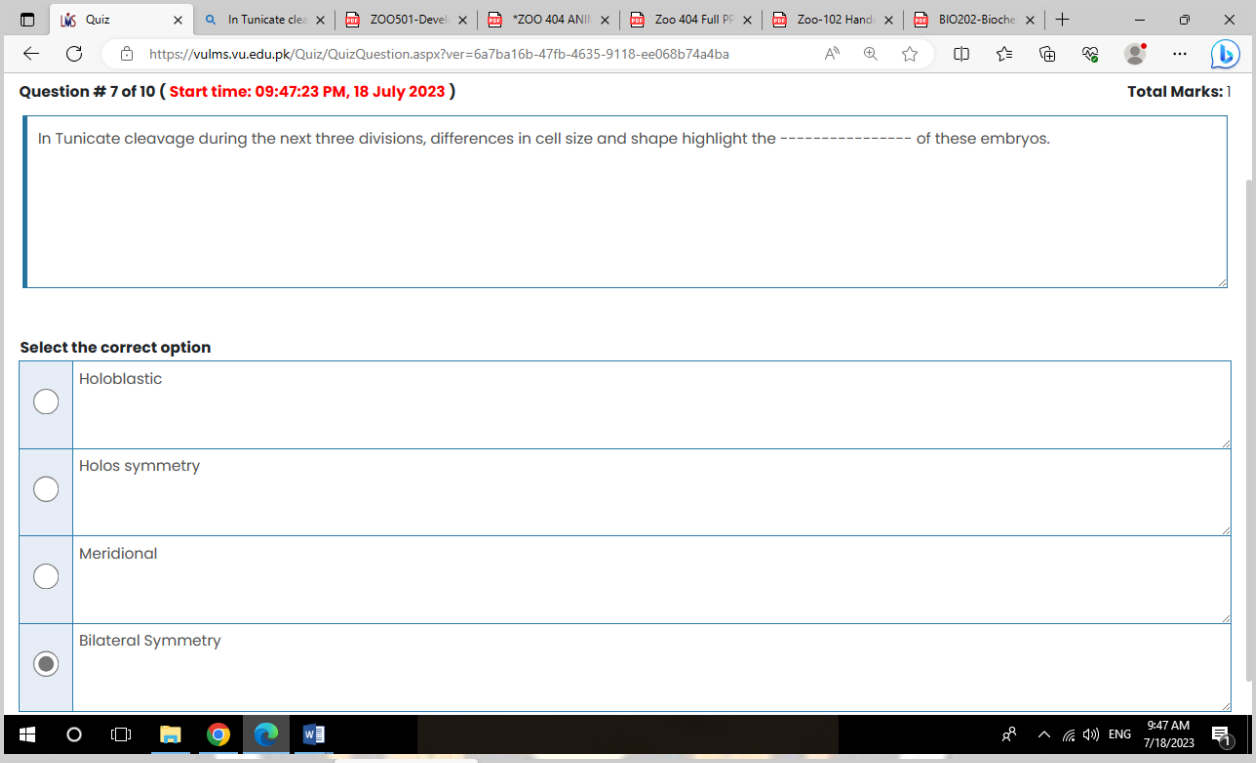
Scanning electron micrographs show beautifully the incomplete nature of discoidal meroblastic cleavage in fish eggs.



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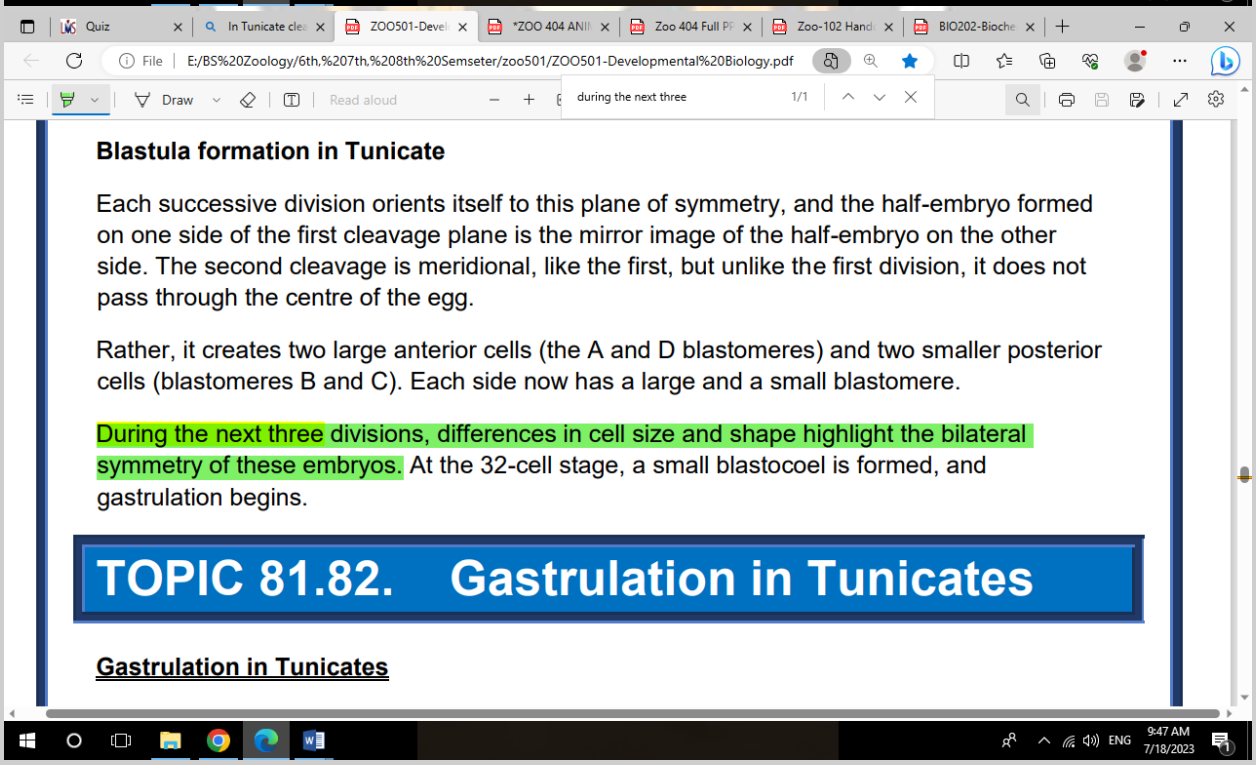


Question # 7 of 10 (Start time: 09:47:23 PM, 18 July 2023) Total Marks: 1

In Tunicate cleavage during the next three divisions, differences in cell size and shape highlight the ---------- of these embryos.

Select the correct option

- Holoblastic
- Holos symmetry
- Meridional
- Bilateral Symmetry



Blastula formation in Tunicate

Each successive division orients itself to this plane of symmetry, and the half-embryo formed on one side of the first cleavage plane is the mirror image of the half-embryo on the other side. The second cleavage is meridional, like the first, but unlike the first division, it does not pass through the centre of the egg.

Rather, it creates two large anterior cells (the A and D blastomeres) and two smaller posterior cells (blastomeres B and C). Each side now has a large and a small blastomere.

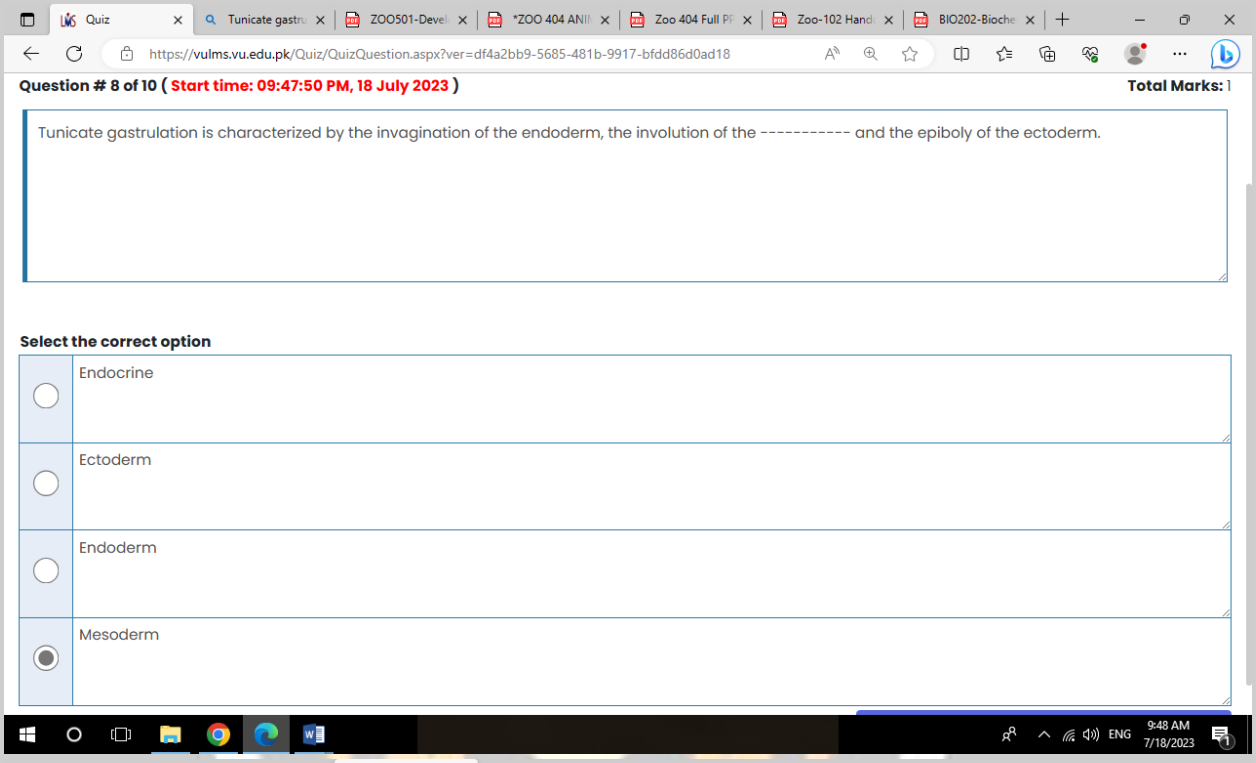
During the next three divisions, differences in cell size and shape highlight the bilateral symmetry of these embryos. At the 32-cell stage, a small blastocoel is formed, and gastrulation begins.

TOPIC 81.82. Gastrulation in Tunicates

Gastrulation in Tunicates

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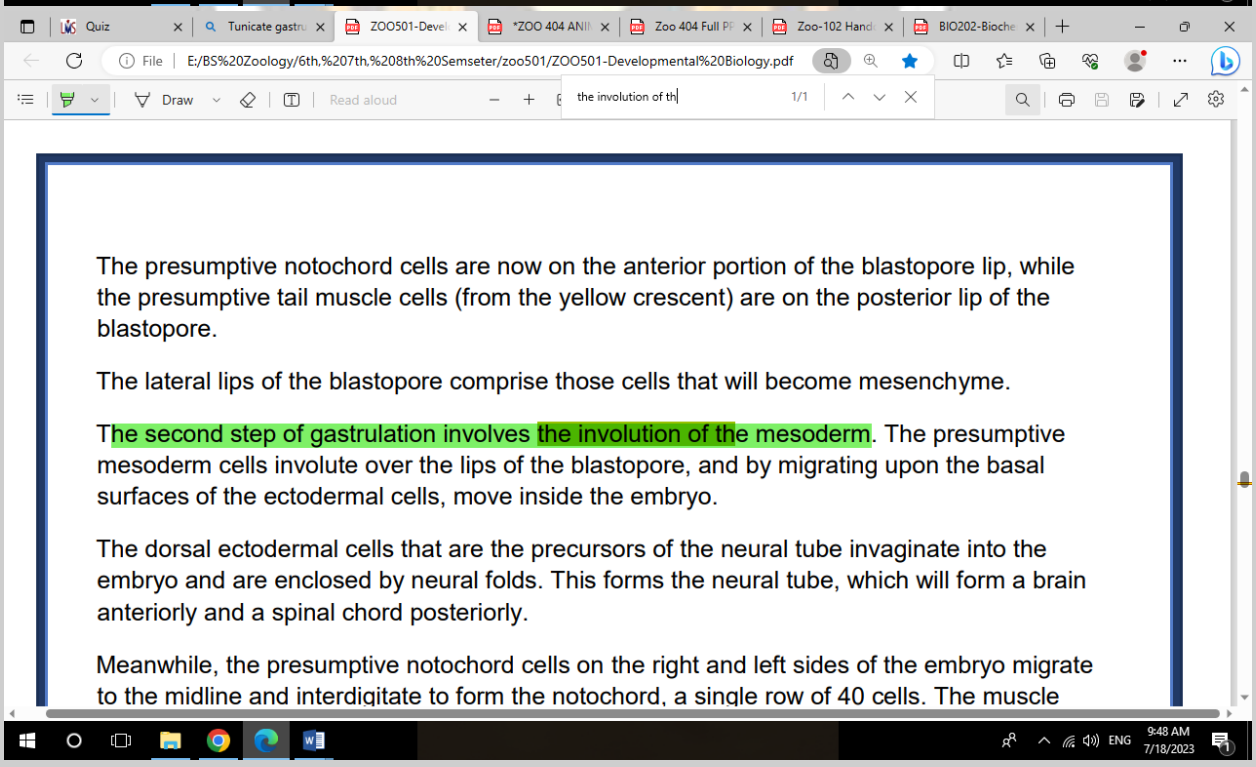


Question # 8 of 10 (Start time: 09:47:50 PM, 18 July 2023) Total Marks: 1

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the ---------- and the epiboly of the ectoderm.

Select the correct option

- Endocrine
- Ectoderm
- Endoderm
- Mesoderm



The presumptive notochord cells are now on the anterior portion of the blastopore lip, while the presumptive tail muscle cells (from the yellow crescent) are on the posterior lip of the blastopore.

The lateral lips of the blastopore comprise those cells that will become mesenchyme.

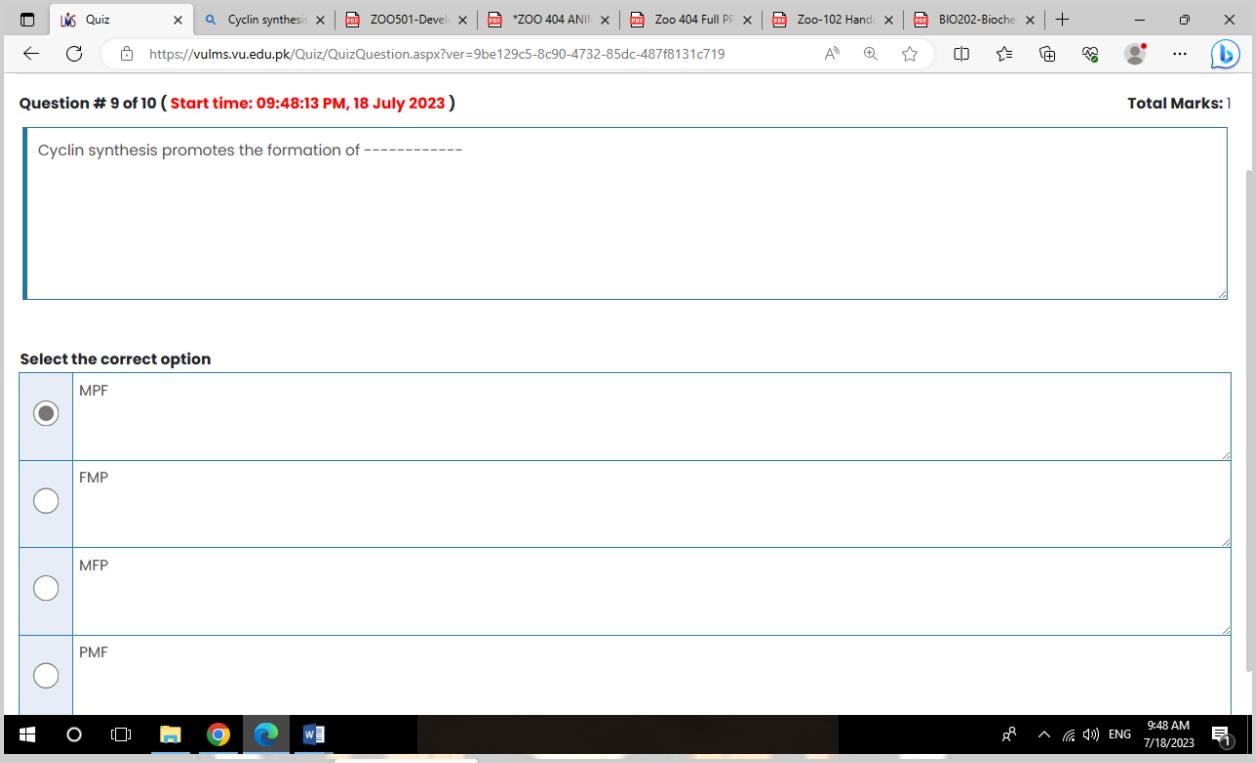
The second step of gastrulation involves the involution of the mesoderm. The presumptive mesoderm cells involute over the lips of the blastopore, and by migrating upon the basal surfaces of the ectodermal cells, move inside the embryo.

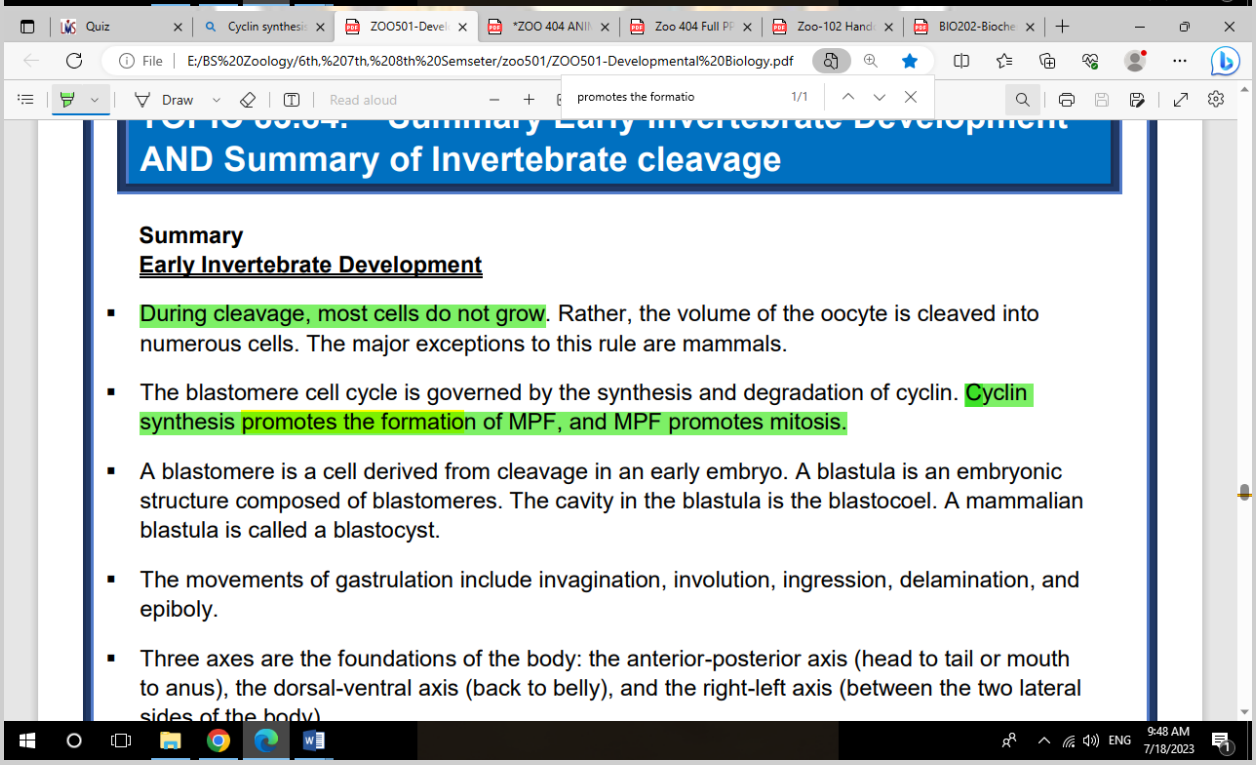
The dorsal ectodermal cells that are the precursors of the neural tube invaginate into the embryo and are enclosed by neural folds. This forms the neural tube, which will form a brain anteriorly and a spinal chord posteriorly.

Meanwhile, the presumptive notochord cells on the right and left sides of the embryo migrate to the midline and interdigitate to form the notochord, a single row of 40 cells. The muscle

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Question # 10 of 10 (Start time: 09:48:50 PM, 18 July 2023) **Total Marks:** 1

In fish eggs, cleavage occurs only in the ---------- a thin region of yolk-free cytoplasm at the animal cap of the egg.

Select the correct option

- (●) Blastodisc
- ( ) Holodisc
- ( ) Neurotodic
- ( ) Mesodisk

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TOPIC 85. 86. Cleavage in Fish

Early Development of Vertebrates

Cleavage in Fishes

In fish eggs, cleavage occurs only in the **blastodisc**, a thin region of yolk-free cytoplasm at the animal cap of the egg. Most of the egg cell is full of yolk. The cell divisions do not completely divide the egg, so this type of cleavage is called **meroblastic.**

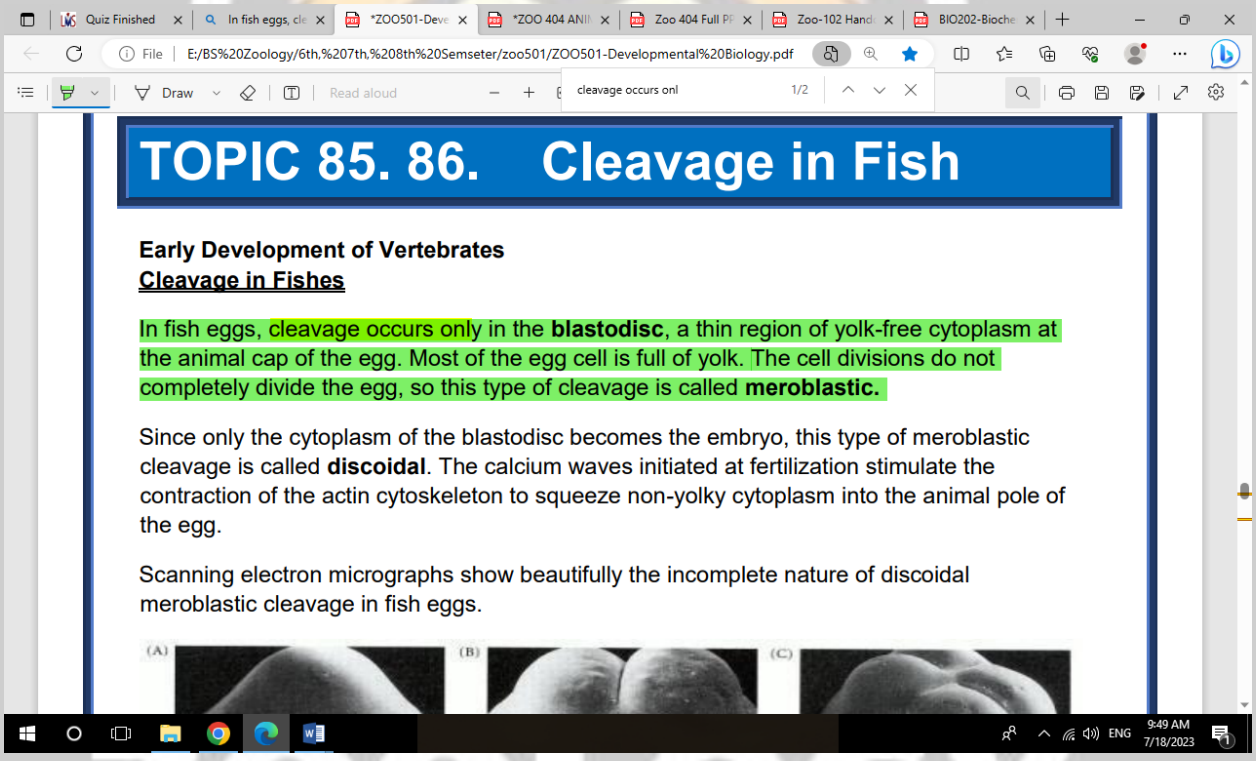
Since only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called **discoidal**. The calcium waves initiated at fertilization stimulate the contraction of the actin cytoskeleton to squeeze non-yolky cytoplasm into the animal pole of the egg.

Scanning electron micrographs show beautifully the incomplete nature of discoidal meroblastic cleavage in fish eggs.



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TOPIC 85. 86. Cleavage in Fish

Early Development of Vertebrates

Cleavage in Fishes

In fish eggs, cleavage occurs only in the **blastodisc**, a thin region of yolk-free cytoplasm at the animal cap of the egg. Most of the egg cell is full of yolk. The cell divisions do not completely divide the egg, so this type of cleavage is called **meroblastic.**

Since only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called **discoidal**. The calcium waves initiated at fertilization stimulate the contraction of the actin cytoskeleton to squeeze non-yolky cytoplasm into the animal pole of the egg.

Scanning electron micrographs show beautifully the incomplete nature of discoidal meroblastic cleavage in fish eggs.

(A) (B) (C)

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Remember me in you Precious Prayer

Jazak Allah

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